India Public Transport Usage Analysis

Project documentation | Tools: Excel, MySQL, Python, Power BI | Dataset: synthetic, 2023 to 2024

1. Project overview

This project analyses public transport usage (Metro and Bus) across six major Indian cities: Delhi, Mumbai, Bengaluru, Chennai, Hyderabad and Kolkata. It covers the full analytics workflow: data preparation, cleaning, SQL analysis, Python analysis and an interactive Power BI dashboard.

Important note: the dataset is synthetic. It was simulated to resemble Indian city transit patterns (weekday and weekend behaviour, holidays, weather effects, metro vs bus growth). It is not official data, so findings describe the simulated patterns and should not be presented as real-world statistics.

2. Objectives and business questions

1. Which cities and modes carry the most riders, and how do they compare?
2. How did ridership grow from 2023 to 2024 for Metro and Bus?
3. How do weekends, holidays and weather (rain, heavy rain, fog, heat) change ridership and punctuality?
4. Which city and mode earns the most revenue per rider?
5. How efficiently is the fleet used (riders per vehicle)?
6. What share of total public transport riders does the Metro carry in each city?

3. Dataset

- **Rows:** 8,772 (6 cities x 2 modes x 731 days)
- **Period:** 1 January 2023 to 31 December 2024 (daily)
- **Grain:** one row per city, mode and date
- **Missing values:** about 1% blanks in `temperature_c` and `on_time_pct`, left in on purpose to practise cleaning

Data dictionary

Column	Type	Description
record_id	Integer	Unique row ID
city	Text	Delhi, Mumbai, Bengaluru, Chennai, Hyderabad, Kolkata
mode	Text	Metro or Bus
date	Date	Day of the record
year, month	Integer	Calendar year and month number
day_of_week	Text	Monday to Sunday
is_weekend	0/1	1 if Saturday or Sunday
is_holiday	0/1	1 on listed public holidays
weather	Text	Clear, Hot, Fog, Rain, Heavy Rain
temperature_c	Decimal	Daily temperature (blanks present)
avg_fare_inr	Decimal	Average fare per trip in INR
vehicles_in_service	Integer	Trains (Metro) or buses (Bus) in service
daily_ridership	Integer	Passengers carried that day
avg_trip_distance_km	Decimal	Average trip distance
peak_hour_share_pct	Decimal	Share of riders travelling in peak hours
on_time_pct	Decimal	Punctuality percentage (blanks present)
women_riders_pct	Decimal	Share of women riders
daily_revenue_inr	Integer	Fare revenue for the day

4. Tools and workflow

Step	Tool	Output
1. Data preparation and cleaning	Excel	Clean table, helper columns, pivot tables
2. SQL analysis	MySQL 8	Table, import, cleaning, 14 analysis queries
3. Statistical analysis and charts	Python (pandas, matplotlib)	10 charts and a cleaned CSV
4. Data model and dashboard	Power BI	4-page interactive dashboard
5. Documentation	This document	Project write-up

5. Data cleaning

1. Converted the data to an Excel table and checked data types (date, numbers, text).
2. Checked duplicates on `record_id` (none expected).
3. Checked unique values for city, mode and weather for spelling or spacing problems.
4. Filled blank `temperature_c` and `on_time_pct` values with the average for the same city and mode (`AVERAGEIFS` in Excel, `UPDATE ... JOIN` in MySQL, `groupby().transform('mean')` in pandas).
5. Validated ranges: percentages must not exceed 100 and ridership must not be negative.
6. Added helper columns: quarter, season, day type (Weekday, Weekend, Holiday), revenue per rider, riders per vehicle.

6. Data model (Power BI)

A star schema was used.

Table	Type	Key columns
Fact_Transport	Fact	date, city, mode, weather, plus all numeric measures columns
Dim_Date	Dimension	Date, Year, Month, Quarter, Weekday, Day Type, Season
Dim_City	Dimension	city, state, region
Dim_Mode	Dimension	mode, mode_type (Rail or Road)
Dim_Weather	Dimension	weather, weather_group (Dry or Wet)

Relationships are one-to-many from each dimension to the fact table, single direction. `Dim_Date` is marked as the date table. State, region, mode type and weather group were added manually because they are not in the raw data.

7. SQL analysis (MySQL)

The SQL script creates the database and table, imports the CSV (blank cells become NULL), runs quality checks, fills NULL values and answers the business questions with queries on:

- Average daily ridership by city and mode
- Year-over-year growth (2023 vs 2024) using a CTE
- Weekday vs weekend and holiday impact
- Weather impact on ridership and on-time performance
- Monthly trend and cumulative revenue (window functions)
- Revenue per rider, fleet utilisation and metro share by city
- City ranking using `RANK()`

8. Python analysis

The Python script loads and cleans the data, creates features, calculates summary tables and saves 10 charts:

1. Monthly metro ridership trend by city (line)
2. City vs mode average ridership (grouped bar)

3. Year-over-year growth (horizontal bar)
4. Day type index, weekday = 100 (bar)
5. Weather effect, % vs clear day (bar)
6. Seasonality heat map, month by city
7. Revenue per rider (bar)
8. Bus on-time % by weather (box plot)
9. Mode share of total ridership (stacked bar)
10. Correlation matrix for Metro (heat map)

9. Key DAX measures

Measure	Purpose
Total Riders, Avg Daily Riders	Core ridership metrics
Total Revenue, Revenue per Rider	Revenue metrics
Riders per Vehicle, Revenue per Vehicle	Fleet efficiency
Metro Share %	Metro riders divided by total riders
Riders PY, YoY Growth %	Year-over-year comparison using `SAMEPERIODLASTYEAR`
Riders Change	Current minus previous year (waterfall chart)
Cumulative Revenue	Running total of revenue over time
Weekend vs Weekday %, Rain Impact %	Behaviour impact
City Rank	`RANKX` ranking of cities

Only two calculated columns were created (Day Type and Season). Everything else is a measure, because measures respond to slicers and filters.

10. Dashboard structure

Each page has slicers (City, Mode, Year) at the top, KPI cards below, then charts, with Metro in teal and Bus in amber.

Page	Purpose	Main visuals
Overview	Big-picture summary	KPI cards, city vs mode bar, donut share, ranking table
Growth	Trends and growth	Monthly line chart, this year vs previous year, YoY matrix, waterfall by city and mode
Behaviour	Weekday, holiday and weather impact	Day type column, weather bars, city by month heat map
Revenue and Fleet Efficiency	Money and fleet use	Cumulative revenue area chart, revenue per rider, scatter of fleet vs ridership, riders per vehicle, detail table

Navigation uses page navigator buttons, and Metro / Bus / All bookmarks switch the mode view.

11. Key findings

Fill this section from your own dashboard numbers. Replace each bracket with the exact figure you see.

- **Ridership:** [City] has the highest Metro ridership at [number] riders per day. Bus carries more riders than Metro in [list cities].
- **Growth:** Metro ridership grew [x]% from 2023 to 2024, compared with [y]% for Bus.
- **Day type:** Weekend Metro ridership is [x]% lower than weekdays, and holidays are [y]% lower.
- **Weather:** Heavy rain reduces Bus ridership by [x]% and Metro ridership by [y]% compared with clear days.
- **Revenue:** [City and mode] earns the highest revenue per rider at INR [number].
- **Fleet use:** [City] has the highest riders per vehicle at [number].